

Sea of Okhotsk pelagic reconstruction departures

This record explains the adopted reconstruction of the 2000–2014 Okhotsk Sea epipelagic food web and every numerical departure from the article's generalized Figure 9, Table 3 and the supporting feeding tables. It distinguishes published values, human readings, computational conversions and scientific assumptions so that the researcher can judge plausibility and uncertainty.

Adopted model ID: 52_GM2019_Fig9_Pelagic_balanced_(2000-2014). [Native model](#). The model is a reconstruction rather than an author Ecopath export. The 22 source groups are retained together with the runtime's external-food import group.

Primary evidence: [Gorbatenko and Melnikov 2019](#), Table 3 printed page 154 / PDF page 12 and Figure 9 printed page 157 / PDF page 15. Supporting quantitative diets and chemical factors: [Gorbatenko dissertation 2018](#), Tables 4.11, 4.13–4.16, 4.52 and 7.1. The [source feeding audit](#) preserves recovered cells, scope differences and conflicts.

The exact native file reloads with main balance True. All 21 living groups have zero biomass accumulation and finite nonnegative respiration, egestion and other mortality. GE, TE and With Egestion return WARN rather than an unqualified scientific pass. Native file SHA256: 9beff5a4e525dcf4253bf83472253c7a4a9f7e7db10379ecef64eb33cd32a656.

Adopted assumptions and their consequences

These changes complete a generalized production-flow scheme as a native mass-balance input. Preserved production values constrain the repairs, but balance alone does not establish author intent. Every table value below is rounded for reading; linked JSON ledgers retain the exact values.

[Microbial stock and split consistency](#) records the wet-stock increase 64 → 65.067 million tonnes (+1.067 million tonnes; +1.6672%). Copepod bacterial wet-food share rises from 0.652777778 to 0.666666667; protozoan EE falls from 0.957892397 to 0.932837989 under the declared 0.95 availability bound. This deliberate food allocation change is not an author measurement.

Change	Article or baseline	Adopted value	Evidence uncertainty and consequence
Common currency and area	Source wet stocks and million tC/yr diagram flows	Native tC/km ² stocks and tC/km ² /yr flows; effective area 1,544,000 km ²	The area is implied by primary production 694.8 million tC/yr and 450 gC/m ² /yr. Prey-specific wet/C conversion is required; the area is not measured author GIS.
Table 3 carbon-stock header	The printed stock-column heading includes per year	Treat stock as million tonnes C, without yr ⁻¹	Biomass and body conversion establish a stock; production and PB carry the time dimension. Original header is preserved, correction is declared.
Table 7.1 intermediate factor headings	Intermediate headings suggest carbon/dry ratios	Use the numerical dry/wet intermediate fractions and independently coherent final wet/C factors	The source Other row remains discrepant. Proxy factors for absent prey are assumptions rather than a correction that proves every row.
Microbial stock split	Pooled carbon B 6.8 million tC; wet B 64 million t; PB 20; separate B unknown	Bacteria B 5.055; Protozoa B 1.755 million tC; PB 20 for both; implied wet B 65.067 million t	Production-proportional split preserves P 101.1 + 35.1 = 136.2. Carbon B differs by 0.01 million tC; wet B rises by 1.067 million t. Separate stocks/rates and protozoan chemistry are assumptions.
Protozoa wet/C	No unambiguous group-specific measurement	10 wet tonnes per tonne C	Explicit best guess. Dissertation page 60 contextual ratio 270/27 is ambiguous because the sentence literally says bacteria.
Catch and migration	Not complete source Ecopath inputs	Catch, immigration and emigration 0 in every group	Authorized closed migration / zero-catch reconstruction. It does not establish regional catch sustainability or absence of actual animal movements.
Living BA	Unknown/free in source reconstruction	0 in all 21 living groups	Sustainable-stock best guess under zero catch. This avoids depletion as a numerical repair; it is not measured annual accumulation.

Change	Article or baseline	Adopted value	Evidence uncertainty and consequence
GS	Source carbon GS unknown; inherited handling could insert 0.2	Unknown in input; actual LIM solves 0.1–0.35 across 20 consumers	Standard GS bounds 0.10–0.35. The source calorie assimilation 0.70 is not an observed carbon GS 0.30.
Mortality and unassimilated routing	Fate of these flows unspecified	All M0 and U to sole Detritus pool	Explicit detrital recycling boundary assumption; no alternate sink or export.
Detritus stock and accumulation	Detritus B unknown; consumption 310 million tC/yr	Assumed B 310/20 = 15.5 million tC; BA 374.43368 million tC/yr	The stock assumes 18.25-day feeding-inventory turnover. Residual BA is 24.16 times assumed inventory each year under zero import/export and all M0/U routing. This large nonsteady accumulation is not measured storage, burial or a stationary detritus stock.
External food	Figure-only import 0; unallocated source food remains	13.357614 million tC/yr across native imports; synthetic group 23	Food categories absent from the 22 groups and unknown residual food are retained, with declared proxy factors and allocations. Their origins and donor-web PPR remain uncertain.
Diet completion	Figure-only arrows and incomplete prose fractions	Full source wet totals retained; exact carbon DC derived from completed prey flows	Unknown food is explicitly imported, not dropped by renormalizing a partial diet. Runtime normalization is disabled; raw and loaded DC totals agree.
Figure F62 and F67	F62 disputed 0.023 or 0.025; F67 formerly unread	F62 0.023 preferred; 0.025 tested. F67 0.05	Accepted human readings. They do not resolve tentative arrow routes. Values are million tC/yr.
Figure F77	Earlier endpoint prey 19	Prey 15 → consumer 20, 0.01 million tC/yr	Source-path correction accepted independently of balancing. Source raster and prior extraction remain available.
Synthetic runtime import group	No author biological compartment	Native helper group 23 with dummy B = 1 tC/km ² supplies imported food	Runtime P and Q equal import flux and are computational completion values, not observed biological rates. They are excluded from living source-group counts and source production sums.
Predatory fish wet/C	Table 3 alignment assigns 5.43 wet/C to this small pooled fish group	5.43 wet tonnes per tonne C	The source row association is tentative; chemistry transfers also affect catch taxa mapped to this weak pelagic analogue.
EE and zero apex rows	Source EE unavailable	EE derived from D/P with BA 0; groups 17,18,20,21 EE 0	Finite physical bounds are checked. TE assigns these apex groups zero SPPR despite consumption; that software convention is a material limitation.

Twelve assumed arrow magnitude corrections

The following source labels are divided by 10 or 100. Their source readings remain unchanged in the preserved extraction. Production-availability closure at zero catch, migration and living BA motivates the choices. No independent source identifies these decimals as errors, and three routes are visually clear. These are Very low confidence scientific hypotheses.

Rejected alternatives include leaving all arrow magnitudes unchanged, using substantial negative stock accumulation, changing verified production boxes, reversing clear arrow heads, and dropping source groups. The source-fed unrepaired case is algebraically balanced but has negative mortality or EE above 1; the accepted case avoids those failures. Sensitivity does not turn the selected decimal hypothesis into source proof.

Arrow	Prey to consumer	Source million tC/yr	Adopted million tC/yr	Route evidence
F41	Herring (10) → Pollock 30-60 cm (15)	0.685	0.00685	tentative
F42	Herring (10) → Pollock 30-60 cm (15)	0.139	0.0139	tentative
F44	Deep-sea smelt (11) → Pollock 30-60 cm (15)	0.114	0.0114	clear
F52	Capelin (13) → Pollock 30-60 cm (15)	0.57	0.0057	tentative
F53	Capelin (13) → Squid IV (16)	0.114	0.00114	clear
F57	Capelin (13) → Pollock 30-60 cm (15)	0.024	0.0024	tentative
F64	Pollock greater than 60 cm (19) → Predatory salmon (17)	0.057	0.0057	tentative

Arrow	Prey to consumer	Source million tC/yr	Adopted million tC/yr	Route evidence
F65	Capelin (13) → Predatory fish (20)	0.057	0.0057	tentative
F66	Squid III (9) → Squid IV (16)	0.588	0.00588	tentative
F70	Squid III (9) → Predatory fish (20)	0.042	0.0042	tentative
F71	Herring (10) → Predatory fish (20)	0.2	0.02	tentative
F75	Squid IV (16) → Predatory fish (20)	0.11	0.011	clear

Source and adopted stocks production intake and rates

B, P and Q in this table use million tonnes C and years. Source B is the rounded carbon column; adopted B generally derives directly from wet B divided by the primary body factor. Figure production P is held exactly for every living group. PB is derived from that P and adopted B rather than forcing rounded Table 3 stock, production and PB to be simultaneously exact. Producer/detritus runtime self-throughput is distinct from biological food intake.

Group	Source B C	Adopted B C	Figure P C/yr	Figure Q C/yr	Adopted Q C/yr	Source PB	Native PB
Phytoplankton (1)	3.2	3.16	694.8	not food intake	0	220	219.873
Bacteria (2)	absent	5.055	101.1	310	310	absent	20
Protozoa (3)	absent	1.755	35.1	108	108	absent	20
Copepods (4)	7.7	7.73571	117.2	176	182.864	15.2	15.1505
Euphausiids (5)	5.6	5.53774	56.1	87	87.9686	10.1	10.1305
Hyperiid (6)	0.41	0.406897	4.752	4.383	9.71027	11.7	11.6786
Chaetognaths (7)	2.13	2.135	12.78	20.666	23.1172	6	5.98595
Salmon (8)	0.09	0.097561	0.105	0.0731	0.134922	1.14	1.07625
Squid III (9)	0.04	0.0420168	0.105	1.69381	1.93157	2.5	2.499
Herring (10)	0.35	0.347222	0.119	2.756	3.0315	0.34	0.34272
Deep-sea smelt (11)	0.21	0.206897	0.06	0.4615	0.845504	0.29	0.29
Pollock less than 30 cm (12)	0.1	0.0784314	0.197	1.305	1.30573	2.5	2.51175
Capelin (13)	0.09	0.0909091	0.03	1.055	1.31393	0.35	0.33
Jellyfish (14)	0.01	0.0143759	0.144	0.0505	0.234404	10	10.0168
Pollock 30-60 cm (15)	0.767	0.766877	0.59	6.0421	4.55035	0.7	0.769354
Squid IV (16)	0.042	0.041876	0.1	0.816	1.03493	2.5	2.388
Predatory salmon (17)	0.002	0.00243902	0.002	0.0833	0.032	0.84	0.82
Baleen whales (18)	0.133	0.132781	0.001	0.21415	0.21415	0.01	0.00753121
Pollock greater than 60 cm (19)	0.045	0.0450045	0.009	0.06984	0.06984	0.2	0.19998
Predatory fish (20)	0.022	0.0220994	0.004	0.4874	0.1193	0.2	0.181
Predatory mammals (21)	0.045	0.0449355	0.001	0.1042	0.1042	0.03	0.0222541
Detritus (22)	absent	15.5	absent	not food intake	0	absent	44.157

Recovering the three failed energy budgets

Hyperiid retain source wet Q 152.5 million tonnes/yr, including copepods 56.7 and Other 95.8. The retained clear euphausiid flow 0.283 million tC/yr accounts for 2.9998 million wet tonnes within Other. The remaining 92.8002 million wet tonnes are assumed 40% chaetognaths, 50% absent larvae/tunicates and 10% gelatinous food. The latter two use wet/C proxies 13.3 and 285.2. This supplies sufficient carbon intake while retaining Figure P 4.752; the split is unmeasured and alternative residual mixtures remain plausible.

Salmon source Table 4.52 supplies residence-integrated total wet Q 1.79694 million tonnes/yr. The 1.7968 leaf sum and 0.00014 residual are both retained. Amphipods map to hyperiid; gelatinous food maps to jellyfish; fish food maps to smelt. Squid prey splits equally between stages using equal source wet stocks. Pteropods, decapods and Oikopleura enter

imports with recorded factors. Table 4.54's apparent copepod/amphipod swap is rejected in favor of direct species feeding Table 4.52. These aggregations and proxy factors are assumptions.

Jellyfish complete wet Q 3.1771 million tonnes/yr derives from 2006–2014 large/small feeding tables rather than the 2000–2014 Figure intake sum 0.0505. The stock-period and size-scope mismatch remains. Copepods, euphausiids, hyperiid/amphipod, chaetognath and cannibalistic jellyfish categories use represented prey; absent prey categories and Other use imports. Jellyfish P 0.144 and primary body factor 285.2 remain unchanged. Prey-specific factors explain why wet jellyfish P 41 can exceed wet intake without the same carbon-energy contradiction.

Group	Preserved P million tC/yr	Figure Q	Adopted Q	Solved GS	Final R tC/km ² /yr
Hyperiiids (6)	4.752	4.383	9.710272	0.1	2.5824124
Salmon (8)	0.105	0.0731	0.13492214	0.13420893	0.007651806
Jellyfish (14)	0.144	0.0505	0.23440358	0.1	0.04336996

Every source feeding component and its conversion

The table retains every wet quantity used to complete source diets. All quantities are million wet tonnes/yr or million tC/yr. Import means the prey is not represented internally. The basis records the source scope, residual/stage allocation and chemical proxy. Figure paths without a replaced diet remain in the adopted carbon-flow matrix. Primary pooled microheterotroph food is split 2/3 bacteria and 1/3 protozoa by wet mass for copepods, a source-informed but unmeasured split; the initial Figure-proportional split failed microbial availability and is retained as a rejected trial. Euphausiids use the Figure-relative wet split. The squid source 64% zooplankton and 36% nekton are retained in aggregate; unknown plankton splits 90/10 and unknown nekton 20/80 between stages III/IV using their stated feeding roles. Small pollock's approximately 0.2% category-scale adjustment and all residual imports are explicit below.

[Exact full food component ledger](#) | [Final carbon flow matrix](#)

Jellyfish source assimilation 70% is expressed in calorie terms and cannot establish carbon GS = 0.30. The actual free native LIM GS is 0.10. Squid IV obtains 88.3064% of its adopted carbon food from imports; its unallocated nekton and assumed 20/80 stage split materially weaken inference about local donor production.

Consumer	Prey or external category	Wet food	Wet per C	Carbon food	Source or assumption
6 Hyperiiids	4 Copepods	56.7	14	4.05	2019 p. 153 and dissertation Table 4.11
6 Hyperiiids	5 Euphausiids	2.9998	10.6	0.283	Clear Figure F13 nested inside source Other
6 Hyperiiids	7 Other: chaetognaths	37.1201	20	1.856	Assumed 40% of residual; qualitative feeding evidence
6 Hyperiiids	23 Other: absent larval crustaceans and tunicates	46.4001	13.3	3.48873	Assumed 50%; primary Table 3 Other plankton factor
6 Hyperiiids	14 Other: gelatinous food	9.28002	285.2	0.0325386	Assumed 10%; jellyfish primary factor
8 Salmon	5 Euphausiids	0.4719	10.6	0.0445189	Dissertation Table 4.52
8 Salmon	6 Amphipods represented as hyperiiids	0.8678	14.5	0.0598483	Table 4.52; taxonomic lump assumption
8 Salmon	4 Copepods	0.0513	14	0.00366429	Table 4.52 direct species sums; conflicting Table 4.54 rejected
8 Salmon	7 Chaetognaths	0.0605	20	0.003025	Table 4.52
8 Salmon	23 Pteropods	0.0768	17.79	0.00431703	Table 4.52 wet; Table 7.1 warm season factor
8 Salmon	23 Decapods	0.0191	13.3	0.00143609	Table 4.52; Other planktonfactor proxy
8 Salmon	23 Oikopleura	0.0612	20.97	0.00291845	Table 4.52; Table 7.1 warm season factor
8 Salmon	14 Gelatinous organisms	0.0448	285.2	0.000157083	Table 4.52; jellyfish lump assumption
8 Salmon	9 Squid III	0.02345	11.9	0.00197059	Table 4.52; equal wetstock stage split
8 Salmon	16 Squid IV	0.02345	11.94	0.00196399	Table 4.52; equal wetstock stage split
8 Salmon	11 Fish represented as smelt	0.0965	8.7	0.011092	Table 4.52; Figure F36 prey analogue assumption
8 Salmon	23 Rounded total-minus-leaf discrepancy	0.00014	13.3	1.05263e-05	1.79694 total minus 1.7968 leafsum retained

Consumer	Prey or external category	Wet food	Wet per C	Carbon food	Source or assumption
14 Jellyfish	5 Euphausiids	0.3322	10.6	0.0313396	Dissertation Tables 4.15/4.16; primary prey factor or declared external proxy
14 Jellyfish	4 Copepods	2.4773	14	0.17695	Dissertation Tables 4.15/4.16; primary prey factor or declared external proxy
14 Jellyfish	6 Amphipods	0.0367	14.5	0.00253103	Dissertation Tables 4.15/4.16; primary prey factor or declared external proxy
14 Jellyfish	7 Chaetognaths	0.0259	20	0.001295	Dissertation Tables 4.15/4.16; primary prey factor or declared external proxy
14 Jellyfish	23 Pteropods	0.0058	18.5626	0.000312456	Dissertation Tables 4.15/4.16; primary prey factor or declared external proxy
14 Jellyfish	23 Mysids	0.0057	13.7191	0.00041548	Dissertation Tables 4.15/4.16; primary prey factor or declared external proxy
14 Jellyfish	23 Decapod larvae	0.0048	13.3	0.000360902	Dissertation Tables 4.15/4.16; primary prey factor or declared external proxy
14 Jellyfish	14 Jellyfish	0.0328	285.2	0.000115007	Dissertation Tables 4.15/4.16; primary prey factor or declared external proxy
14 Jellyfish	23 Fish eggs	0.0629	10.02	0.00627745	Dissertation Tables 4.15/4.16; primary prey factor or declared external proxy
14 Jellyfish	23 Fish larvae	0.012	10.02	0.0011976	Dissertation Tables 4.15/4.16; primary prey factor or declared external proxy
14 Jellyfish	23 Meroplankton	0.0639	13.3	0.00480451	Dissertation Tables 4.15/4.16; primary prey factor or declared external proxy
14 Jellyfish	23 Other	0.1171	13.3	0.00880451	Dissertation Tables 4.15/4.16; primary prey factor or declared external proxy
4 Copepods	1 Phytoplankton	2261.76	20	113.088	2019 p. 153 copepods
4 Copepods	2 Bacteria part of pooled microheterotrophs	422.117	9.4	44.906	2019 p. 153 copepods; assumed 2/3 pooled wet share; Figure 2:1 carbon prior and EE headroom
4 Copepods	3 Protozoa part of pooled microheterotrophs	211.058	10	21.1058	2019 p. 153 copepods; assumed 1/3 pooled wet share; Figure 2:1 carbon prior and EE headroom
4 Copepods	23 Unallocated source 1.7% other food	50.065	13.3	3.76429	2019 p. 153 copepods; explicit Other plankton import
5 Euphausiids	1 Phytoplankton	1140.58	20	57.0289	2019 p. 153 euphausiids
5 Euphausiids	2 Bacteria part of pooled microheterotrophs	164.079	9.4	17.4552	2019 p. 153 euphausiids; Figure wet-relative split
5 Euphausiids	3 Protozoa part of pooled microheterotrophs	116.368	10	11.6368	2019 p. 153 euphausiids; Figure wet-relative split
5 Euphausiids	23 Unallocated source 1.7% other food	24.5752	13.3	1.84776	2019 p. 153 euphausiids; explicit Other plankton import
7 Chaetognaths	4 Copepods	281.4	14	20.1	2019 p. 153 direct annual amount; avoids rounded 88% conflict
7 Chaetognaths	5 Euphausiids	5.9996	10.6	0.566	Retain clear Figure F16; not source 2.2% ambiguous denominator
7 Chaetognaths	23 Unallocated other food	32.6004	13.3	2.45116	2019 total 320 minus known food; imported source unknown
11 Deep-sea smelt	4 Copepods	4.18803	14	0.299145	2019 p. 155
11 Deep-sea smelt	5 Euphausiids	4.74369	10.6	0.447518	2019 p. 155
11 Deep-sea smelt	6 Hyperiid	0.12325	14.5	0.0085	Retain Figure F32 inside 13% remainder
11 Deep-sea smelt	7 Chaetognaths	0.1	20	0.005	Retain Figure F33 inside 13% remainder
11 Deep-sea smelt	23 Unallocated other food	1.13503	13.3	0.0853406	Explicit remainder; absent prey import
10 Herring	4 Copepods	15.4	14	1.1	2019 p. 155 herring; Figure within-zooplankton split
10 Herring	5 Euphausiids	15.9	10.6	1.5	2019 p. 155 herring; Figure within-zooplankton split
10 Herring	6 Hyperiid	2.262	14.5	0.156	2019 p. 155 herring; Figure within-zooplankton split

Consumer	Prey or external category	Wet food	Wet per C	Carbon food	Source or assumption
10 Herring	23 Other zooplankton	2.3417	13.3	0.176068	2019 p. 155 herring; preserved unallocated zooplankton
10 Herring	23 Absent nekton food	0.9963	10.02	0.0994311	2019 p. 155 herring; primary Other nekton factor proxy
13 Capelin	5 Euphausiids	10.6	10.6	1	2019 p. 156 capelin; Figure within-zooplankton split
13 Capelin	6 Hyperiids	0.7975	14.5	0.055	2019 p. 156 capelin; Figure within-zooplankton split
13 Capelin	23 Other zooplankton	3.03296	13.3	0.228042	2019 p. 156 capelin; preserved unallocated zooplankton
13 Capelin	23 Absent nekton food	0.30954	10.02	0.0308922	2019 p. 156 capelin; primary Other nekton factor proxy
12 Pollock less than 30 cm	4 Copepods	10.2828	14	0.734483	2019 p. 156 small pollock; Figure within-zooplankton split scaled 0.20% to printed subtotal
12 Pollock less than 30 cm	5 Euphausiids	5.31023	10.6	0.500965	2019 p. 156 small pollock; Figure within-zooplankton split scaled 0.20% to printed subtotal
12 Pollock less than 30 cm	6 Hyperiids	0.665625	14.5	0.0459052	2019 p. 156 small pollock; Figure within-zooplankton split scaled 0.20% to printed subtotal
12 Pollock less than 30 cm	7 Chaetognaths	0.0598763	20	0.00299382	2019 p. 156 small pollock; Figure within-zooplankton split scaled 0.20% to printed subtotal
12 Pollock less than 30 cm	8 Salmon	0.1476	8.2	0.018	2019 p. 156 small pollock; represented nonplankton
12 Pollock less than 30 cm	23 Absent nekton food	0.0339	10.02	0.00338323	2019 p. 156 small pollock; primary Other nekton factor proxy
9 Squid III	5 Euphausiids	14.84	10.6	1.4	Retained Figure path after explicit decimal scenario
9 Squid III	6 Hyperiids	3.799	14.5	0.262	Retained Figure path after explicit decimal scenario
9 Squid III	7 Chaetognaths	0.3	20	0.015	Retained Figure path after explicit decimal scenario
9 Squid III	8 Salmon	0.01968	8.2	0.0024	Retained Figure path after explicit decimal scenario
9 Squid III	12 Pollock less than 30 cm	0.0306	10.2	0.003	Retained Figure path after explicit decimal scenario
9 Squid III	13 Capelin	0.10032	8.8	0.0114	Retained Figure path after explicit decimal scenario
9 Squid III	14 Jellyfish	0.002852	285.2	1e-05	Retained Figure path after explicit decimal scenario
9 Squid III	23 Absent other plankton	0.127033	13.3	0.00955137	2019 aggregate 64%;90/10 residual stage allocation
9 Squid III	23 Absent nekton prey	2.28668	10.02	0.228211	2019 aggregate 36%;20/80 residual stage allocation; III planktivory/IV nekton feeding prose
16 Squid IV	6 Hyperiids	1.653	14.5	0.114	Retained Figure path after explicit decimal scenario
16 Squid IV	9 Squid III	0.069972	11.9	0.00588	Retained Figure path after explicit decimal scenario
16 Squid IV	13 Capelin	0.010032	8.8	0.00114	Retained Figure path after explicit decimal scenario
16 Squid IV	23 Absent other plankton	0.0141148	13.3	0.00106126	2019 aggregate 64%;90/10 residual stage allocation
16 Squid IV	23 Absent nekton prey	9.14672	10.02	0.912846	2019 aggregate 36%;20/80 residual stage allocation; III planktivory/IV nekton feeding prose

Solved physical budgets and scientific limits

All flows below are tC/km²/yr except EE and GS. The engine's final main model is reloaded from the persisted file and balanced. No failed LIM fallback values or legacy migration-based balanced copy are used. Living BA is 0, detritus BA is positive, all catch/migration values are 0, and all U/M0 flows enter detritus. Detritus residual accumulation of 374.43368 million tC/yr is 24.16 times its assumed 15.5 million tC inventory per year. With zero export/import and all M0/U routed to detritus, this is a large nonsteady residual, not measured burial/storage or a stationary stock. Solver values are a consequence of the selected food and stock assumptions, not independently measured author parameters.

The 12 decimal changes resolve source predation greater than production for squid III, herring, smelt, capelin, squid IV and large pollock. This prevents negative mortality/depletion but materially changes internal predation. Extensive external food preserves total consumer intake; it weakens inference about which local production supports consumers. Scientific validation must assess donor food sources, chemistry, allocation and temporal scope rather than infer validity from algebraic closure.

Group	EE	GS	BA	R	U	M0
Phytoplankton (1)	0.352788	0	0	0	0	291.246
Bacteria (2)	0.943236	0.100304	0	115.159	20.1387	3.71684
Protozoa (3)	0.932838	0.100472	0	40.1872	7.02781	1.5268
Copepods (4)	0.225804	0.1	0	30.6851	11.8435	58.7667
Euphausiids (5)	0.18399	0.1	0	14.9429	5.69745	29.6491
Hyperiid (6)	0.163002	0.1	0	2.58241	0.628904	2.57605
Chaetognaths (7)	0.1549	0.1	0	5.19783	1.49723	6.99507
Salmon (8)	0.310476	0.134209	0	0.00765181	0.0117278	0.0468912
Squid III (9)	0.707148	0.140756	0	1.00693	0.176088	0.0199154
Herring (10)	0.536891	0.143004	0	1.60556	0.280775	0.035693
Deep-sea smelt (11)	0.667199	0.138284	0	0.433021	0.0757253	0.0129327
Pollock less than 30 cm (12)	0.0797462	0.12639	0	0.611204	0.106885	0.117416
Capelin (13)	0.911333	0.145448	0	0.707788	0.123776	0.0017228
Jellyfish (14)	0.232088	0.1	0	0.04337	0.0151816	0.0716187
Pollock 30-60 cm (15)	0.186441	0.129547	0	2.1832	0.381791	0.310881
Squid IV (16)	0.60964	0.134465	0	0.515393	0.0901302	0.0252824
Predatory salmon (17)	0	0.35	0	0.0121762	0.00725389	0.00129534
Baleen whales (18)	0	0.148152	0	0.117502	0.0205484	0.000647668
Pollock greater than 60 cm (19)	0.633333	0.35	0	0.0235725	0.0158316	0.00213731
Predatory fish (20)	0	0.143856	0	0.0635609	0.0111153	0.00259067
Predatory mammals (21)	0	0.147418	0	0.0568905	0.00994883	0.000647668
Detritus (22)	1	0	242.509	0	0	0

Mandatory baselines and sensitivity outcomes

Every case uses the actual native runtime. Main balance is an algebraic engine result; physical admissibility independently checks flow signs, EE/GS bounds and the accepted boundaries. Thus the unrepaired text-fed case is rejected even though its main balanced flag is True. F62 = 0.025 gives the same pass/fail classification as the preferred 0.023 in all four structural cases. Raw failed inputs and returns remain available.

The complete F62 = 0.025 native model balances and all three direct methods remain WARN with finite coefficients. Maximum relative all-source SPPR changes are 2.7% for GE, 3.0% for TE and 2.7% for With Egestion. These maxima cover native groups; the ambiguous route remains tentative.

[Full sensitivity returns and summary](#) | [Every group and source-scope comparison](#)

Case	F62	Main balance	Physical admissibility	Unsolved groups
figure only	0.023	False	False	6, 8, 14
figure only	0.025	False	False	6, 8, 14

Case	F62	Main balance	Physical admissibility	Unsolved groups
text plus figure unrepaired	0.023	True	False	none
text plus figure unrepaired	0.025	True	False	none
reduced figure source repairs	0.023	True	True	none
reduced figure source repairs	0.025	True	True	none
text plus figure adopted	0.023	True	True	none
text plus figure adopted	0.025	True	True	none

Why algebraic balance alone fails the unrepaired case

At preferred F62 = 0.023, the source-fed unrepaired case returns main balance True but the following six groups require negative other mortality and EE above 1. The selected decimal assumptions remove these violations. EE is dimensionless; M0 is tC/km²/yr. These actual returned failures motivate a repair but do not identify author intent.

[All eight case inputs and physical violations](#)

Group	Unrepaired EE	Adopted EE	Unrepaired M0	Adopted M0
Squid III (9)	6.61115	0.707148	-0.381587	0.0199154
Herring (10)	8.7995	0.536891	-0.601127	0.035693
Deep-sea smelt (11)	2.3772	0.667199	-0.0535181	0.0129327
Capelin (13)	25.9133	0.911333	-0.484067	0.0017228
Squid IV (16)	1.59964	0.60964	-0.0388368	0.0252824
Pollock greater than 60 cm (19)	6.33333	0.633333	-0.0310881	0.00213731

Numerical alternatives to the decimal assumptions

Preserving unrepaired outgoing predation would require these production or stock/PB multipliers under the declared 95% production availability bound. These large changes are rejected because independently confirmed production boxes are retained. They are nevertheless energy-feasible given the completed food and therefore illustrate non-uniqueness. The selected arrow counts are minimum only within the discrete decimal-reduction search, not proof of a source error or a globally minimal ecological repair. Herring has a different feasible minimum-exponent decimal allocation (F41/10, F42/10 and F71/100), so the adopted allocation remains a choice.

[Exact alternative calculations and restricted search](#)

Group	Source P	Unrepaired predation	Adopted predation	Required multiplier	Alternative P	Changed labels selected/min
Squid III (9)	0.105	0.694171	0.0742506	6.9591	0.730706	2 / 2
Herring (10)	0.119	1.04914	0.06589	9.28032	1.10436	3 / 3
Deep-sea smelt (11)	0.06	0.142632	0.040032	2.50231	0.150139	1 / 1
Capelin (13)	0.03	0.7774	0.02734	27.2772	0.818316	4 / 4
Squid IV (16)	0.1	0.159964	0.060964	1.68383	0.168383	1 / 1
Pollock greater than 60 cm (19)	0.009	0.057	0.0057	6.66667	0.06	1 / 1

Import fidelity and unusual production efficiencies

The eight EwE tables and companion files retain 1,510 cells with all numeric and missing-state values within tolerance (maximum numeric difference 4×10^{-14}). Strings are not all exact: normalization of labels is documented. The raw database converter omits nonfeeding fate/import structure and fails native admission with a singular matrix. An explicit constructor-admission copy restores documented schema fields, structural zeros and the native companion; its actual roundtrip state agrees within 10^{-12} . The converter output is not presented as an independently balanced native model.

The interface checker returns 0 errors and 1 warning for 20 blank GS inputs. The indicative mass-balance check returns 0 errors, 8 warnings and 1 note. The database converter reports 20 INDETERMINATE groups because unknown GS causes its indicative default 0.2; this does not replace the actual native LIM result with freely solved GS.

The independent screen flags P/Q beyond its usual 0.02–0.5 range: copepods 0.641, euphausiids 0.638, chaetognaths 0.553, salmon 0.778 and jellyfish 0.614, with baleen whales 0.00467 and predatory mammals 0.00960 below it. These ratios remain scientific physiology or aggregation concerns despite nonnegative energy budgets. The producer has only B and PB as known author-style inputs; native production closure and software import sufficiency are distinct checks.

[Actual import tool returns](#) | [Detailed database roundtrip](#) | [Source fidelity checks](#)

Source error hypotheses examined

The earlier jellyfish source audit examined the following 16 explanations. The final completed diet adds declared assumptions where that audit could only retain missing food; it does not retroactively establish an exact author allocation or chemical factor.

Hypothesis	Source evidence and arithmetic	Interpretation
J01 Wet and carbon production conflict	$4.10 \times 10 = 41.0$ million wet tonnes/yr; $41.0 / 285.20 = 0.14375877$ million tC/yr, rounding to 0.144.	Rejected. Primary production is internally coherent.
J02 Rounded carbon biomass	$4.10 / 285.20 = 0.01437588$ million tC, printed as 0.01; multiplying the rounded stock by PB 10 gives only 0.10.	Supported rounding explanation. Derived stock and exact printed stock remain distinct.
J03 Table 3 stock header	The carbon-biomass header contains a per-year unit even though wet stock divided by body factor reproduces the cells.	Interpret as carbon stock, with a source header error. No numerical correction follows.
J04 Tenfold production reduction	A hypothetical $0.144 \rightarrow 0.0144$, wet P 41 $\rightarrow$ 4.1 or PB 10 $\rightarrow$ 1.0 would ease the diagram intake failure. Table 3 and Figure 9 independently repeat the original values.	Rejected as a source-supported correction. Balance improvement alone does not identify a typo.
J05 Figure box is consumption	Figure 9 defines box numbers as production; Table 3 identifies 0.144 as carbon production.	Rejected role swap.
J06 Arrow units are wet mass	Figure 9 specifies million tC/yr for arrow consumption. Treating the three tentative jellyfish inflows as wet gives only 0.00359224 million tC/yr.	Rejected currency reinterpretation.
J07 Consumer body factor for food	$3.1771 / 285.2 = 0.01114060$ million tC/yr would use jellyfish tissue chemistry for copepods and euphausiids.	Rejected. Convert each prey with the prey's factor.
J08 Missing represented food	Three tentative Figure inflows sum to 0.0505 million tC/yr. Supporting copepods 2.4773, euphausiids 0.3322 and chaetognaths 0.0259 million wet tonnes/yr alone convert to 0.20958462 million tC/yr.	Supported source discrepancy. Supporting diets can repair nominal energy balance without changing production; their period differs.
J09 F48 omitted zero	Hypothetical $0.046 \rightarrow 0.46$ would give diagram Q 0.4645 million tC/yr. The printed label is 0.046; route is tentative.	Unsupported author-error hypothesis. Preserve as a rejected/sensitivity option rather than an author correction.
J10 F24 or F50 powers of ten	$F50 \times 10$ gives Q 0.0775; $F24 \times 10$ gives 0.0640; $F24 \times 100$ gives 0.199 million tC/yr.	No independent correction evidence. Exact source labels remain available.
J11 Reverse jellyfish arrows	Reversing 0.0006 or 0.00001 inflows would raise 0.0505 only to 0.0511 or 0.05051, still below P 0.144. The 0.0006 head visibly points toward predatory salmon.	Insufficient repair; do not reverse a clear arrow to force balance.
J12 Wrong tentative endpoints	F24, F48 and F50 all have tentative endpoint readings among crossing red curves.	Plausible routing uncertainty. Source feeding tables supply a separate, explicit replacement diet.
J13 Supporting jellyfish total	Prose gives large-jelly Q 772.8 kt; Table 4.15 and its exclusive rows give 772.7 kt. With small jellyfish 2404.4 kt, table total is 3177.1 kt.	Use table sums; preserve 0.1 kt conflict. It cannot explain the large diagram deficit.
J14 Exact full carbon food	Supporting tables close the full wet diet, but several prey categories lack directly applicable chemical factors or stage allocations.	Exact author carbon total is unavailable. Final residual conversions/allocation must be declared scientific assumptions.
J15 Table 7.1 intermediate heading	All 18 intermediate cells match dry/wet fraction rounded to two decimals. For example, copepod 0.146×0.4275 gives factor 16.02179, matching printed 16.02.	Likely heading/content mismatch. Retain coherent final factors; preserve the distinct Other-row inconsistency.

Hypothesis	Source evidence and arithmetic	Interpretation
J16 Energy assimilation equals carbon GS	The dissertation assumes calorie assimilation 0.70; its complement is 0.30. It supplies no independently measured carbon GS.	Do not label carbon GS 0.30 as recovered measurement. Standard native LIM solves GS within its bounds in this reconstruction.

Every native scalar input change

This appendix enumerates every changed scalar or input flag relative to the preserved human-reading native model. Unknown means the input sentinel -9999; absent means no original value. Baseline biomass is wet t/km²; final biomass is tC/km². Biomass and habitat biomass repeat the same full-area stock, while PB/QB are yr⁻¹ in both. The manuscript policies above explain the scientific basis; this table also retains computational flags and explicitly written zeros. Numeric display uses eight significant digits; exact original/final strings remain in the linked ledger.

Group ID and name	Field	Baseline value	Final input
1 Phytoplankton	biomass	40.932642	2.0466321
1 Phytoplankton	biomass_accum	unknown	0
1 Phytoplankton	biomass_accum_rate	unknown	0
1 Phytoplankton	biomass_habitat_area	40.932642	2.0466321
1 Phytoplankton	detritus_import	unknown	0
1 Phytoplankton	emigration	unknown	0
1 Phytoplankton	emigration_rate	unknown	0
1 Phytoplankton	export	unknown	0
1 Phytoplankton	gs	unknown	0
1 Phytoplankton	immigration	unknown	0
1 Phytoplankton	pb	220	219.87342
2 Bacteria	b_hab_area_input	false	true
2 Bacteria	biomass	unknown	3.2739637
2 Bacteria	biomass_accum	unknown	0
2 Bacteria	biomass_accum_rate	unknown	0
2 Bacteria	biomass_habitat_area	unknown	3.2739637
2 Bacteria	detritus_import	unknown	0
2 Bacteria	emigration	unknown	0
2 Bacteria	emigration_rate	unknown	0
2 Bacteria	export	unknown	0
2 Bacteria	gs	0.2	unknown
2 Bacteria	immigration	unknown	0
2 Bacteria	pb	unknown	20
2 Bacteria	pb_input	false	true
2 Bacteria	qb	unknown	61.32542
2 Bacteria	qb_input	false	true
3 Protozoa	b_hab_area_input	false	true
3 Protozoa	biomass	unknown	1.136658
3 Protozoa	biomass_accum	unknown	0
3 Protozoa	biomass_accum_rate	unknown	0
3 Protozoa	biomass_habitat_area	unknown	1.136658
3 Protozoa	detritus_import	unknown	0
3 Protozoa	emigration	unknown	0
3 Protozoa	emigration_rate	unknown	0
3 Protozoa	export	unknown	0
3 Protozoa	gs	0.2	unknown
3 Protozoa	immigration	unknown	0
3 Protozoa	pb	unknown	20
3 Protozoa	pb_input	false	true
3 Protozoa	qb	unknown	61.538462

Group ID and name	Field	Baseline value	Final input
3 Protozoa	qb_input	false	true
4 Copepods	biomass	70.142487	5.0101776
4 Copepods	biomass_accum	unknown	0
4 Copepods	biomass_accum_rate	unknown	0
4 Copepods	biomass_habitat_area	70.142487	5.0101776
4 Copepods	detritus_import	unknown	0
4 Copepods	diet_imp	0	0.020585149
4 Copepods	emigration	unknown	0
4 Copepods	emigration_rate	unknown	0
4 Copepods	export	unknown	0
4 Copepods	gs	0.2	unknown
4 Copepods	immigration	unknown	0
4 Copepods	pb	15.2	15.150508
4 Copepods	qb	unknown	23.638948
4 Copepods	qb_input	false	true
5 Euphausiids	biomass	38.018135	3.5866165
5 Euphausiids	biomass_accum	unknown	0
5 Euphausiids	biomass_accum_rate	unknown	0
5 Euphausiids	biomass_habitat_area	38.018135	3.5866165
5 Euphausiids	detritus_import	unknown	0
5 Euphausiids	diet_imp	0	0.021004754
5 Euphausiids	emigration	unknown	0
5 Euphausiids	emigration_rate	unknown	0
5 Euphausiids	export	unknown	0
5 Euphausiids	gs	0.2	unknown
5 Euphausiids	immigration	unknown	0
5 Euphausiids	pb	10.1	10.130494
5 Euphausiids	qb	unknown	15.885306
5 Euphausiids	qb_input	false	true
6 Hyperiids	biomass	3.8212435	0.26353404
6 Hyperiids	biomass_accum	unknown	0
6 Hyperiids	biomass_accum_rate	unknown	0
6 Hyperiids	biomass_habitat_area	3.8212435	0.26353404
6 Hyperiids	detritus_import	unknown	0
6 Hyperiids	diet_imp	0	0.35928235
6 Hyperiids	emigration	unknown	0
6 Hyperiids	emigration_rate	unknown	0
6 Hyperiids	export	unknown	0
6 Hyperiids	gs	0.2	unknown
6 Hyperiids	immigration	unknown	0
6 Hyperiids	pb	11.7	11.678644
6 Hyperiids	qb	10.237254	23.864228
7 Chaetognaths	biomass	27.65544	1.382772
7 Chaetognaths	biomass_accum	unknown	0
7 Chaetognaths	biomass_accum_rate	unknown	0
7 Chaetognaths	biomass_habitat_area	27.65544	1.382772
7 Chaetognaths	detritus_import	unknown	0
7 Chaetognaths	diet_imp	0	0.10603197
7 Chaetognaths	emigration	unknown	0
7 Chaetognaths	emigration_rate	unknown	0
7 Chaetognaths	export	unknown	0
7 Chaetognaths	gs	0.2	unknown

Group ID and name	Field	Baseline value	Final input
7 Chaetognaths	immigration	unknown	0
7 Chaetognaths	pb	6	5.9859485
7 Chaetognaths	qb	6.7306698	10.827709
8 Salmon	biomass	0.51813472	0.06318716
8 Salmon	biomass_accum	unknown	0
8 Salmon	biomass_accum_rate	unknown	0
8 Salmon	biomass_habitat_area	0.51813472	0.06318716
8 Salmon	detritus_import	unknown	0
8 Salmon	diet_imp	0	0.064348988
8 Salmon	emigration	unknown	0
8 Salmon	emigration_rate	unknown	0
8 Salmon	export	unknown	0
8 Salmon	gs	0.2	unknown
8 Salmon	immigration	unknown	0
8 Salmon	pb	1.14	1.07625
8 Salmon	qb	1.247425	1.382952
9 Squid III	biomass	0.3238342	0.027212958
9 Squid III	biomass_accum	unknown	0
9 Squid III	biomass_accum_rate	unknown	0
9 Squid III	biomass_habitat_area	0.3238342	0.027212958
9 Squid III	detritus_import	unknown	0
9 Squid III	diet_imp	0	0.12309288
9 Squid III	emigration	unknown	0
9 Squid III	emigration_rate	unknown	0
9 Squid III	export	unknown	0
9 Squid III	gs	0.2	unknown
9 Squid III	immigration	unknown	0
9 Squid III	pb	2.5	2.499
9 Squid III	qb	38.184904	45.971434
10 Herring	biomass	1.619171	0.22488486
10 Herring	biomass_accum	unknown	0
10 Herring	biomass_accum_rate	unknown	0
10 Herring	biomass_habitat_area	1.619171	0.22488486
10 Herring	detritus_import	unknown	0
10 Herring	diet_imp	0	0.090878745
10 Herring	emigration	unknown	0
10 Herring	emigration_rate	unknown	0
10 Herring	export	unknown	0
10 Herring	gs	0.2	unknown
10 Herring	immigration	unknown	0
10 Herring	pb	0.34	0.34272
10 Herring	qb	13.4248	8.7307166
11 Deep-sea smelt	biomass	1.1658031	0.13400036
11 Deep-sea smelt	biomass_accum	unknown	0
11 Deep-sea smelt	biomass_accum_rate	unknown	0
11 Deep-sea smelt	biomass_habitat_area	1.1658031	0.13400036
11 Deep-sea smelt	detritus_import	unknown	0
11 Deep-sea smelt	diet_imp	0	0.10093465
11 Deep-sea smelt	emigration	unknown	0
11 Deep-sea smelt	emigration_rate	unknown	0
11 Deep-sea smelt	export	unknown	0
11 Deep-sea smelt	gs	0.2	unknown

Group ID and name	Field	Baseline value	Final input
11 Deep-sea smelt	immigration	unknown	0
11 Deep-sea smelt	pb	0.29	0.29
11 Deep-sea smelt	qb	2.76225	4.0866004
12 Pollock less than 30 cm	biomass	0.51813472	0.050797521
12 Pollock less than 30 cm	biomass_accum	unknown	0
12 Pollock less than 30 cm	biomass_accum_rate	unknown	0
12 Pollock less than 30 cm	biomass_habitat_area	0.51813472	0.050797521
12 Pollock less than 30 cm	detritus_import	unknown	0
12 Pollock less than 30 cm	diet_imp	0	0.002591065
12 Pollock less than 30 cm	emigration	unknown	0
12 Pollock less than 30 cm	emigration_rate	unknown	0
12 Pollock less than 30 cm	export	unknown	0
12 Pollock less than 30 cm	gs	0.2	unknown
12 Pollock less than 30 cm	immigration	unknown	0
12 Pollock less than 30 cm	pb	2.5	2.51175
12 Pollock less than 30 cm	qb	20.62475	16.648069
13 Capelin	biomass	0.51813472	0.058878945
13 Capelin	biomass_accum	unknown	0
13 Capelin	biomass_accum_rate	unknown	0
13 Capelin	biomass_habitat_area	0.51813472	0.058878945
13 Capelin	detritus_import	unknown	0
13 Capelin	diet_imp	0	0.19706793
13 Capelin	emigration	unknown	0
13 Capelin	emigration_rate	unknown	0
13 Capelin	export	unknown	0
13 Capelin	gs	0.2	unknown
13 Capelin	immigration	unknown	0
13 Capelin	pb	0.35	0.33
13 Capelin	qb	14.246875	14.453278
14 Jellyfish	biomass	2.6554404	0.0093108009
14 Jellyfish	biomass_accum	unknown	0
14 Jellyfish	biomass_accum_rate	unknown	0
14 Jellyfish	biomass_habitat_area	2.6554404	0.0093108009
14 Jellyfish	detritus_import	unknown	0
14 Jellyfish	diet_imp	0	0.094592888
14 Jellyfish	emigration	unknown	0
14 Jellyfish	emigration_rate	unknown	0
14 Jellyfish	export	unknown	0
14 Jellyfish	gs	0.2	unknown
14 Jellyfish	immigration	unknown	0
14 Jellyfish	pb	10	10.01678
14 Jellyfish	qb	0.17636585	16.305341
15 Pollock 30-60 cm	biomass	5.5181347	0.49668179
15 Pollock 30-60 cm	biomass_accum	unknown	0
15 Pollock 30-60 cm	biomass_accum_rate	unknown	0
15 Pollock 30-60 cm	biomass_habitat_area	5.5181347	0.49668179
15 Pollock 30-60 cm	detritus_import	unknown	0
15 Pollock 30-60 cm	emigration	unknown	0
15 Pollock 30-60 cm	emigration_rate	unknown	0
15 Pollock 30-60 cm	export	unknown	0
15 Pollock 30-60 cm	gs	0.2	unknown
15 Pollock 30-60 cm	immigration	unknown	0

Group ID and name	Field	Baseline value	Final input
15 Pollock 30-60 cm	pb	0.7	0.76935446
15 Pollock 30-60 cm	qb	7.1499883	5.9336137
16 Squid IV	biomass	0.3238342	0.027121792
16 Squid IV	biomass_accum	unknown	0
16 Squid IV	biomass_accum_rate	unknown	0
16 Squid IV	biomass_habitat_area	0.3238342	0.027121792
16 Squid IV	detritus_import	unknown	0
16 Squid IV	diet_imp	0	0.88306424
16 Squid IV	emigration	unknown	0
16 Squid IV	emigration_rate	unknown	0
16 Squid IV	export	unknown	0
16 Squid IV	gs	0.2	unknown
16 Squid IV	immigration	unknown	0
16 Squid IV	pb	2.5	2.388
16 Squid IV	qb	19.3068	24.714063
17 Predatory salmon	biomass	0.012953368	0.001579679
17 Predatory salmon	biomass_accum	unknown	0
17 Predatory salmon	biomass_accum_rate	unknown	0
17 Predatory salmon	biomass_habitat_area	0.012953368	0.001579679
17 Predatory salmon	detritus_import	unknown	0
17 Predatory salmon	emigration	unknown	0
17 Predatory salmon	emigration_rate	unknown	0
17 Predatory salmon	export	unknown	0
17 Predatory salmon	gs	0.2	unknown
17 Predatory salmon	immigration	unknown	0
17 Predatory salmon	pb	0.84	0.82
17 Predatory salmon	qb	54.3655	13.12
18 Baleen whales	biomass	0.46696891	0.085997958
18 Baleen whales	biomass_accum	unknown	0
18 Baleen whales	biomass_accum_rate	unknown	0
18 Baleen whales	biomass_habitat_area	0.46696891	0.085997958
18 Baleen whales	detritus_import	unknown	0
18 Baleen whales	emigration	unknown	0
18 Baleen whales	emigration_rate	unknown	0
18 Baleen whales	export	unknown	0
18 Baleen whales	gs	0.2	unknown
18 Baleen whales	immigration	unknown	0
18 Baleen whales	pb	0.01	0.0075312067
18 Baleen whales	qb	3.1297087	1.6128079
19 Pollock greater than 60 cm	biomass	0.3238342	0.029147993
19 Pollock greater than 60 cm	biomass_accum	unknown	0
19 Pollock greater than 60 cm	biomass_accum_rate	unknown	0
19 Pollock greater than 60 cm	biomass_habitat_area	0.3238342	0.029147993
19 Pollock greater than 60 cm	detritus_import	unknown	0
19 Pollock greater than 60 cm	emigration	unknown	0
19 Pollock greater than 60 cm	emigration_rate	unknown	0
19 Pollock greater than 60 cm	export	unknown	0
19 Pollock greater than 60 cm	gs	0.2	unknown
19 Pollock greater than 60 cm	immigration	unknown	0
19 Pollock greater than 60 cm	pb	0.2	0.19998
19 Pollock greater than 60 cm	qb	1.520896	1.5518448
20 Predatory fish	biomass	0.077720207	0.014313114

Group ID and name	Field	Baseline value	Final input
20 Predatory fish	biomass_accum	unknown	0
20 Predatory fish	biomass_accum_rate	unknown	0
20 Predatory fish	biomass_habitat_area	0.077720207	0.014313114
20 Predatory fish	detritus_import	unknown	0
20 Predatory fish	emigration	unknown	0
20 Predatory fish	emigration_rate	unknown	0
20 Predatory fish	export	unknown	0
20 Predatory fish	gs	0.2	unknown
20 Predatory fish	immigration	unknown	0
20 Predatory fish	pb	0.2	0.181
20 Predatory fish	qb	39.929833	5.398325
21 Predatory mammals	biomass	0.15803109	0.029103331
21 Predatory mammals	biomass_accum	unknown	0
21 Predatory mammals	biomass_accum_rate	unknown	0
21 Predatory mammals	biomass_habitat_area	0.15803109	0.029103331
21 Predatory mammals	detritus_import	unknown	0
21 Predatory mammals	emigration	unknown	0
21 Predatory mammals	emigration_rate	unknown	0
21 Predatory mammals	export	unknown	0
21 Predatory mammals	gs	0.2	unknown
21 Predatory mammals	immigration	unknown	0
21 Predatory mammals	pb	0.03	0.022254098
21 Predatory mammals	qb	4.7418852	2.318877
22 Detritus	b_hab_area_input	false	true
22 Detritus	biomass	unknown	10.03886
22 Detritus	biomass_habitat_area	unknown	10.03886
22 Detritus	emigration	unknown	0
22 Detritus	emigration_rate	unknown	0
22 Detritus	gs	unknown	0
22 Detritus	immigration	unknown	0

Every native diet and detritus fate input change

Consumer is the native row group. DC is a dimensionless carbon food fraction; fate values are routing fractions, not feeding quantities. Newly explicit zero entries expand the native representation and remain listed. Source-to-final food quantities and their reasons are above; the exact per-cell basis is in the ledger.

Native row group	Field	Prey or origin ID	Baseline	Final
1 Phytoplankton	diet.proportion	1	absent	0
1 Phytoplankton	diet.detritus_fate	1	absent	0
1 Phytoplankton	diet.proportion	2	absent	0
1 Phytoplankton	diet.detritus_fate	2	absent	0
1 Phytoplankton	diet.proportion	3	absent	0
1 Phytoplankton	diet.detritus_fate	3	absent	0
1 Phytoplankton	diet.proportion	4	absent	0
1 Phytoplankton	diet.detritus_fate	4	absent	0
1 Phytoplankton	diet.proportion	5	absent	0
1 Phytoplankton	diet.detritus_fate	5	absent	0
1 Phytoplankton	diet.proportion	6	absent	0
1 Phytoplankton	diet.detritus_fate	6	absent	0
1 Phytoplankton	diet.proportion	7	absent	0
1 Phytoplankton	diet.detritus_fate	7	absent	0
1 Phytoplankton	diet.proportion	8	absent	0

Native row group	Field	Prey or origin ID	Baseline	Final
1 Phytoplankton	diet.detritus_fate	8	absent	0
1 Phytoplankton	diet.proportion	9	absent	0
1 Phytoplankton	diet.detritus_fate	9	absent	0
1 Phytoplankton	diet.proportion	10	absent	0
1 Phytoplankton	diet.detritus_fate	10	absent	0
1 Phytoplankton	diet.proportion	11	absent	0
1 Phytoplankton	diet.detritus_fate	11	absent	0
1 Phytoplankton	diet.proportion	12	absent	0
1 Phytoplankton	diet.detritus_fate	12	absent	0
1 Phytoplankton	diet.proportion	13	absent	0
1 Phytoplankton	diet.detritus_fate	13	absent	0
1 Phytoplankton	diet.proportion	14	absent	0
1 Phytoplankton	diet.detritus_fate	14	absent	0
1 Phytoplankton	diet.proportion	15	absent	0
1 Phytoplankton	diet.detritus_fate	15	absent	0
1 Phytoplankton	diet.proportion	16	absent	0
1 Phytoplankton	diet.detritus_fate	16	absent	0
1 Phytoplankton	diet.proportion	17	absent	0
1 Phytoplankton	diet.detritus_fate	17	absent	0
1 Phytoplankton	diet.proportion	18	absent	0
1 Phytoplankton	diet.detritus_fate	18	absent	0
1 Phytoplankton	diet.proportion	19	absent	0
1 Phytoplankton	diet.detritus_fate	19	absent	0
1 Phytoplankton	diet.proportion	20	absent	0
1 Phytoplankton	diet.detritus_fate	20	absent	0
1 Phytoplankton	diet.proportion	21	absent	0
1 Phytoplankton	diet.detritus_fate	21	absent	0
3 Protozoa	diet.proportion	1	0.82863772	0.69444444
3 Protozoa	diet.proportion	2	0.17136228	0.30555556
4 Copepods	diet.proportion	1	unknown	0.61842631
4 Copepods	diet.proportion	2	unknown	0.24557044
4 Copepods	diet.proportion	3	unknown	0.1154181
5 Euphausiids	diet.proportion	1	unknown	0.64828701
5 Euphausiids	diet.proportion	2	unknown	0.19842494
5 Euphausiids	diet.proportion	3	unknown	0.13228329
6 Hyperiids	diet.proportion	4	0.95033427	0.4170841
6 Hyperiids	diet.proportion	5	0.049665727	0.029144395
6 Hyperiids	diet.proportion	7	0	0.19113821
6 Hyperiids	diet.proportion	14	0	0.0033509504
7 Chaetognaths	diet.proportion	4	0.97912454	0.86948405
7 Chaetognaths	diet.proportion	5	0.020875464	0.024483979
8 Salmon	diet.proportion	4	0	0.02715852
8 Salmon	diet.proportion	5	0	0.32995968
8 Salmon	diet.proportion	6	0.79914624	0.44357638
8 Salmon	diet.proportion	7	0.050103213	0.022420337
8 Salmon	diet.proportion	9	0.054852997	0.014605373
8 Salmon	diet.proportion	11	0.095897549	0.082210033
8 Salmon	diet.proportion	14	0	0.0011642473
8 Salmon	diet.proportion	16	0	0.014556444
9 Squid III	diet.proportion	5	0.77727052	0.72479792
9 Squid III	diet.proportion	6	0.19897916	0.13564075
9 Squid III	diet.proportion	7	0.015713016	0.007765692

Native row group	Field	Prey or origin ID	Baseline	Final
9 Squid III	diet.proportion	8	0.0010307738	0.0012425107
9 Squid III	diet.proportion	12	0.0016027276	0.0015531384
9 Squid III	diet.proportion	13	0.0052544325	0.0059019259
9 Squid III	diet.proportion	14	0.0001493784	5.177128e-06
10 Herring	diet.proportion	4	0.45885227	0.36285681
10 Herring	diet.proportion	5	0.47375007	0.49480475
10 Herring	diet.proportion	6	0.067397652	0.051459694
11 Deep-sea smelt	diet.proportion	4	0	0.35380692
11 Deep-sea smelt	diet.proportion	5	0.955099	0.52929161
11 Deep-sea smelt	diet.proportion	6	0.024788568	0.010053181
11 Deep-sea smelt	diet.proportion	7	0.020112428	0.0059136359
12 Pollock less than 30 cm	diet.proportion	4	0.62449242	0.56250733
12 Pollock less than 30 cm	diet.proportion	5	0.32250088	0.38366668
12 Pollock less than 30 cm	diet.proportion	6	0.040424732	0.035156708
12 Pollock less than 30 cm	diet.proportion	7	0.0036364077	0.0022928288
12 Pollock less than 30 cm	diet.proportion	8	0.008945563	0.013785383
13 Capelin	diet.proportion	5	0.93002852	0.76107305
13 Capelin	diet.proportion	6	0.069971485	0.041859018
14 Jellyfish	diet.proportion	4	0	0.75489463
14 Jellyfish	diet.proportion	5	0	0.13369942
14 Jellyfish	diet.proportion	6	0.92241737	0.010797764
14 Jellyfish	diet.proportion	7	0.041488038	0.0055246598
14 Jellyfish	diet.proportion	11	0.036094593	0
14 Jellyfish	diet.proportion	14	0	0.00049063677
15 Pollock 30-60 cm	diet.proportion	5	0.76562061	0.96695859
15 Pollock 30-60 cm	diet.proportion	6	0.0032847488	0.0030327337
15 Pollock 30-60 cm	diet.proportion	7	0.031616323	0.021163207
15 Pollock 30-60 cm	diet.proportion	10	0.097390094	0.0045600888
15 Pollock 30-60 cm	diet.proportion	11	0.016280929	0.0025053018
15 Pollock 30-60 cm	diet.proportion	13	0.085807291	0.0017800829
16 Squid IV	diet.proportion	6	0.17123501	0.11015267
16 Squid IV	diet.proportion	9	0.72484306	0.0056815588
16 Squid IV	diet.proportion	13	0.10392193	0.0011015267
17 Predatory salmon	diet.proportion	9	0.13352218	0.38125
17 Predatory salmon	diet.proportion	11	0.048008388	0.1875
17 Predatory salmon	diet.proportion	12	0.0046904747	0.015625
17 Predatory salmon	diet.proportion	13	0.008093368	0.03125
17 Predatory salmon	diet.proportion	14	0.15737922	0.01875
17 Predatory salmon	diet.proportion	16	0.065887373	0.1875
17 Predatory salmon	diet.proportion	19	0.582419	0.178125
18 Baleen whales	diet.proportion	5	0.66234733	0.658417
18 Baleen whales	diet.proportion	9	0.26368036	0.2334812
18 Baleen whales	diet.proportion	10	0.07338734	0.10740135
18 Baleen whales	diet.proportion	11	0.0005397692	0.00065374737
18 Baleen whales	diet.proportion	12	4.5202347e-05	4.6696241e-05
19 Pollock greater than 60 cm	diet.proportion	5	0.10454364	0.10738832
19 Pollock greater than 60 cm	diet.proportion	10	0.0013255344	0.0020045819
19 Pollock greater than 60 cm	diet.proportion	12	0.16364038	0.17468499
19 Pollock greater than 60 cm	diet.proportion	15	0.73049045	0.71592211
20 Predatory fish	diet.proportion	6	0.17249008	0.47778709
20 Predatory fish	diet.proportion	9	0.10430797	0.035205365
20 Predatory fish	diet.proportion	10	0.30052717	0.16764459

Native row group	Field	Prey or origin ID	Baseline	Final
20 Predatory fish	diet.proportion	11	0.020698809	0.095557418
20 Predatory fish	diet.proportion	13	0.10468363	0.047778709
20 Predatory fish	diet.proportion	15	0.023186506	0.083822297
20 Predatory fish	diet.proportion	16	0.27410583	0.092204526
21 Predatory mammals	diet.proportion	8	0.086463501	0.11708253
21 Predatory mammals	diet.proportion	15	0.4801127	0.47984645
21 Predatory mammals	diet.proportion	16	0.4334238	0.40307102
22 Detritus	diet.proportion	1	absent	0
22 Detritus	diet.detritus_fate	1	absent	0
22 Detritus	diet.proportion	2	absent	0
22 Detritus	diet.detritus_fate	2	absent	0
22 Detritus	diet.proportion	3	absent	0
22 Detritus	diet.detritus_fate	3	absent	0
22 Detritus	diet.proportion	4	absent	0
22 Detritus	diet.detritus_fate	4	absent	0
22 Detritus	diet.proportion	5	absent	0
22 Detritus	diet.detritus_fate	5	absent	0
22 Detritus	diet.proportion	6	absent	0
22 Detritus	diet.detritus_fate	6	absent	0
22 Detritus	diet.proportion	7	absent	0
22 Detritus	diet.detritus_fate	7	absent	0
22 Detritus	diet.proportion	8	absent	0
22 Detritus	diet.detritus_fate	8	absent	0
22 Detritus	diet.proportion	9	absent	0
22 Detritus	diet.detritus_fate	9	absent	0
22 Detritus	diet.proportion	10	absent	0
22 Detritus	diet.detritus_fate	10	absent	0
22 Detritus	diet.proportion	11	absent	0
22 Detritus	diet.detritus_fate	11	absent	0
22 Detritus	diet.proportion	12	absent	0
22 Detritus	diet.detritus_fate	12	absent	0
22 Detritus	diet.proportion	13	absent	0
22 Detritus	diet.detritus_fate	13	absent	0
22 Detritus	diet.proportion	14	absent	0
22 Detritus	diet.detritus_fate	14	absent	0
22 Detritus	diet.proportion	15	absent	0
22 Detritus	diet.detritus_fate	15	absent	0
22 Detritus	diet.proportion	16	absent	0
22 Detritus	diet.detritus_fate	16	absent	0
22 Detritus	diet.proportion	17	absent	0
22 Detritus	diet.detritus_fate	17	absent	0
22 Detritus	diet.proportion	18	absent	0
22 Detritus	diet.detritus_fate	18	absent	0
22 Detritus	diet.proportion	19	absent	0
22 Detritus	diet.detritus_fate	19	absent	0
22 Detritus	diet.proportion	20	absent	0
22 Detritus	diet.detritus_fate	20	absent	0
22 Detritus	diet.proportion	21	absent	0
22 Detritus	diet.detritus_fate	21	absent	0
22 Detritus	diet.proportion	22	absent	0
22 Detritus	diet.detritus_fate	22	absent	1

Evidence and reproducibility

[Exact source to final ledger](#) records all 448 changed input cells, all 21 unchanged production boxes, accepted readings, case outcomes and uncertainties. [Actual native reload result](#) and [Runtime settings](#) identify the exact constructor and returned balances. [Independent final gate](#) records an independent reload and physical checks.

The runtime constructor is PPRCalculator with underdetermined=True, zero_catch=True, zero_biomass_accum=False, default_gs=False, weight_flow=1, weight_guess=1, normalize_DC=False and balance_BA_after_DC_normalization=False. GS is cleared in the native input. Engine code hashes and actual returned values are retained with that input identity.

[GE full return](#) | [TE full return](#) | [With Egestion full return](#) | [Independent source matrix inspection](#)

All three direct methods return WARN. Their living networks converge and source matrices contain no negative entries. Four apex groups have EE = 0; TE assigns zero source coefficients to them. Their fractionally assigned 2019 landings total 280,181.78 wet t (8.0%), with independent classic PPR 36,151,868.06 tC (15%). 66 taxa are wholly TE-zero, totaling 275,238.14 wet t (7.9%) and 35,970,559.96 tC (14%) of classic PPR. Numerical coefficient coverage of 99.7604625% does not validate ecological catch footprints. Native zero-catch footprint is separate from observed regional catch. Annual/map use is provisional; production eligibility is false and Monte Carlo/global checks are NOT_RUN.

[Regional unit bridge and per-taxon TE exposure](#)

Regional currency bridge: saved wet-equivalent SPPR = native carbon SPPR × 9 / the assigned group's source wet/C factor. Regional/map PPR then multiply wet catch and divide by 9 once. [Exact carbon-to-regional audit](#)

[New validation record](#) | [All taxon mappings and Sources](#)

[Complete native evidence inventory](#) | [Fresh reload consistency](#)

The researcher review decision remains unsigned. The original article, accepted human-reading variant and prior researcher-edited validation are preserved.